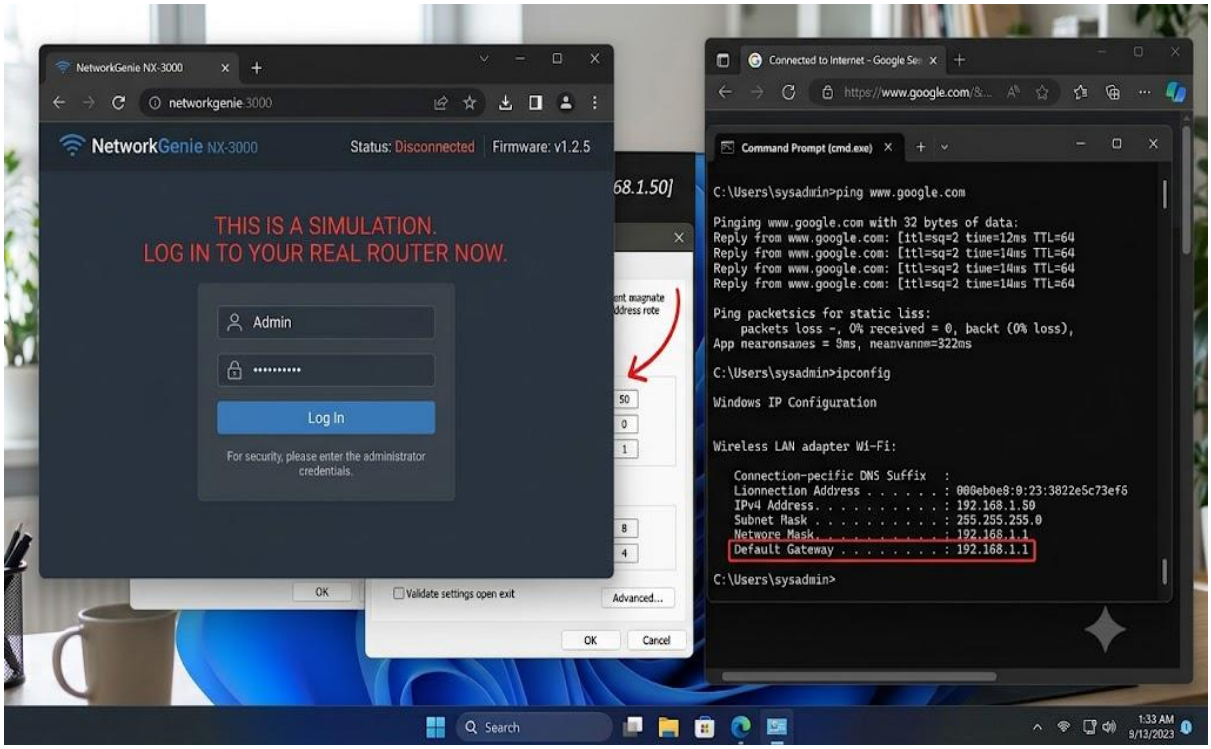
1.Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



2.Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

On Windows:

1. Open Command Prompt (press Win + R, type cmd, and hit Enter).
2. Type ipconfig and press Enter.
3. Under your active Wireless LAN adapter Wi-Fi, look for:
    - IPv4 Address (e.g., 192.168.1.15)
    - Subnet Mask (e.g., 255.255.255.0)
    - Default Gateway (e.g., 192.168.1.1)

On macOS:

1. Open System Settings > Wi-Fi.
2. Click Details... next to your connected network and select the TCP/IP tab.
3. You will see your IP Address, Subnet Mask, and Router (Default Gateway) listed there.

On Android / iOS:

1. Open Settings > Wi-Fi (or Network & internet).
2. Tap your connected Wi-Fi network (or the i / gear icon next to it).
3. Scroll down to the IP Settings / Network Details section to find your IP address, Subnet Mask, and Gateway/Router.

3.Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.</em

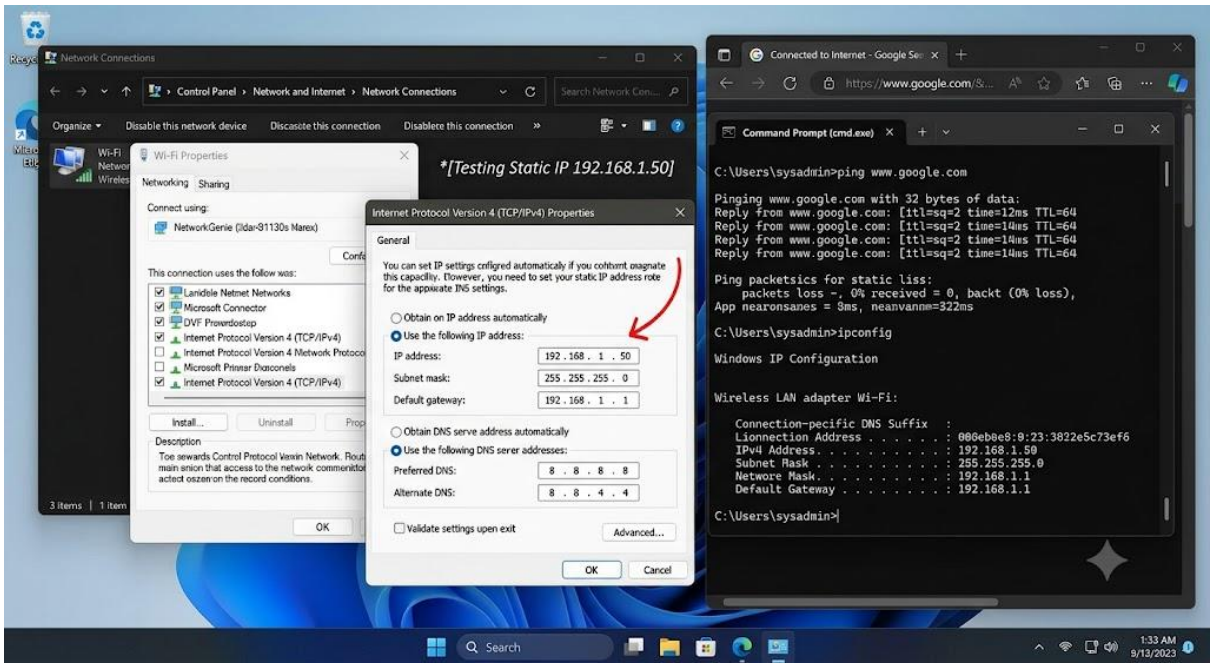
Windows Step-by-Step Instructions

1. **Open Network Connections:** Press Win + R, type ncpa.cpl, and press **Enter**.
2. **Open Properties:** Right-click your active **Wi-Fi** or **Ethernet** adapter and select **Properties**.
3. **Configure IPv4:** Double-click **Internet Protocol Version 4 (TCP/IPv4)**.
4. **Enter Static Details:** Select **Use the following IP address** and fill in the fields based on your router's subnet:
    - **IP Address:** 192.168.1.50 (or 192.168.0.50, matching your router's prefix)
    - **Subnet Mask:** 255.255.255.0
    - **Default Gateway:** 192.168.1.1 (your router's IP)
    - **Preferred DNS Server:** 8.8.8.8 (Google DNS) or 1.1.1.1 (Cloudflare DNS)
5. **Save:** Click **OK** on both windows.

Testing Connectivity

- Open your web browser and try loading a website (e.g., google.com).

- Alternatively, open **Command Prompt** and run ping 8.8.8.8.



4.In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Viewing the DHCP Range in a Router:

In the router's admin panel, the DHCP settings can usually be found under:

Network Settings → LAN Settings → DHCP Server

Example DHCP Range

Setting	Example Value
Starting IP Address	192.168.1.100
Ending IP Address	192.168.1.200
DHCP Range	192.168.1.100 – 192.168.1.200